TOPIC 11 | EVIDENCE REPORT

Database Design for Automation - n8n workflow simulation

Evidence date: 23 October 2026 | Workflow: Topic 11 - Database Design for Automation

Purpose: Demonstrate customer/order/order-item staging, credit-limit validation, success and rejection routing, and audit outputs using n8n Code and IF nodes.

1. Implementation scope

This is an **in-memory simulation**. The screenshots demonstrate logical validation and routing, not actual SQLite/PostgreSQL/MySQL inserts, durable persistence, SQL JOIN execution, or ACID commit/rollback.

2. Execution summary

Scenario	Observed outcome	Result
CUST-001: balance 1,200; limit 5,000; order 1,275	COMMITTED; audit TRANSACTION_VERIFIED; new calculated balance 2,475	PASS
CUST-002: balance 450; limit 500; order 1,275	ROLLED_BACK; reason Credit limit exceeded	PASS
Line items: 900 + 375	Order total and calculated item total both 1,275	PASS

3. Evidence 01 - Complete workflow

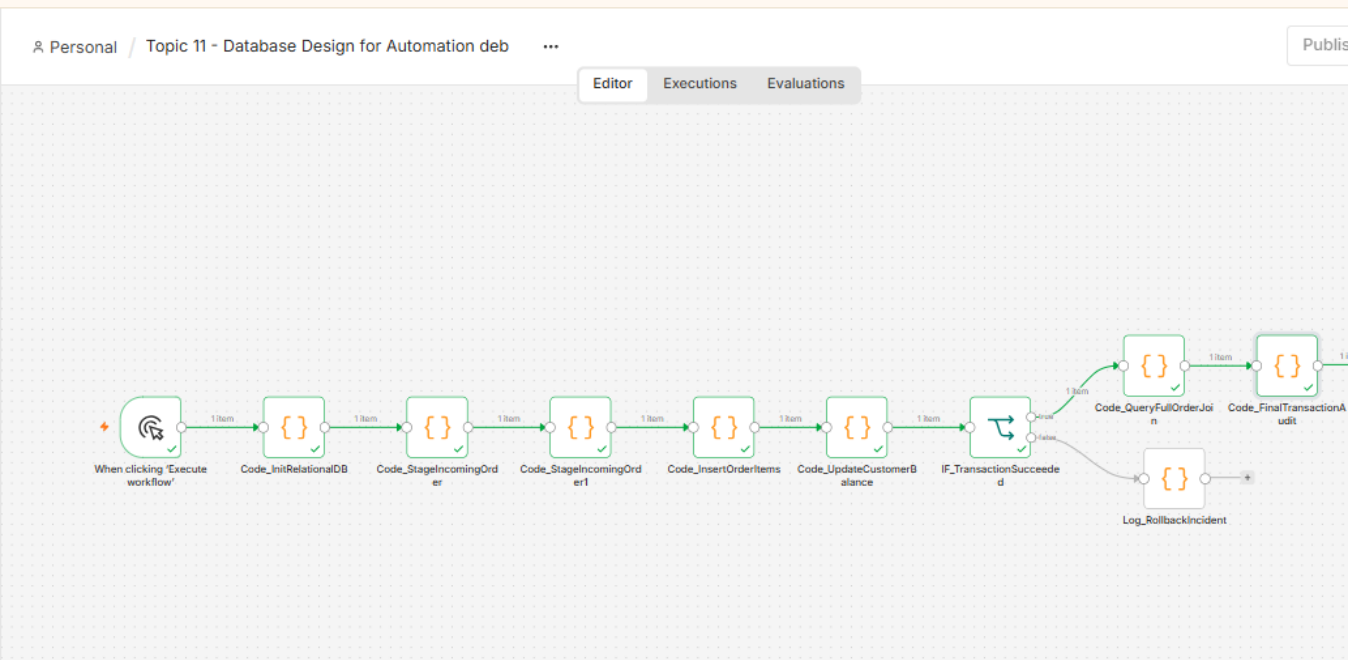


Figure 1. Workflow canvas showing initialization, order staging, order header and item preparation, customer balance check, IF routing, success join simulation, final audit, and rejection logging. Green checks indicate the successful branch ran in the captured execution.

4. Evidence 02 - Successful transaction audit

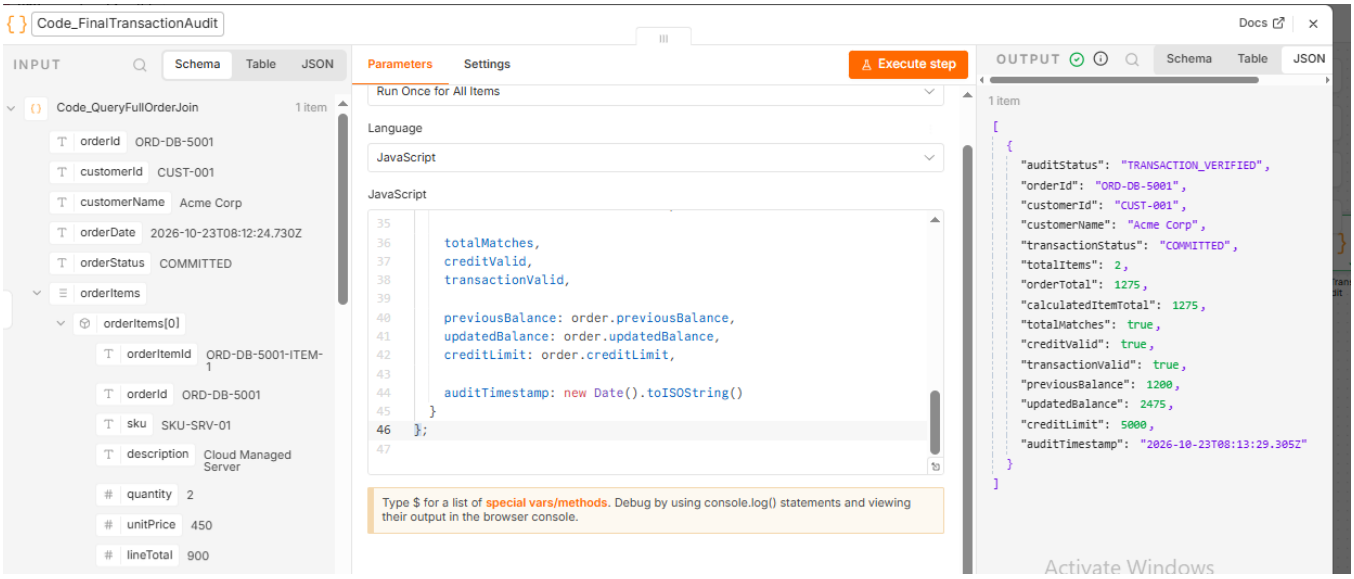


Figure 2. Final audit JSON for CUST-001 reports TRANSACTION_VERIFIED, COMMITTED, 2 items, orderTotal 1275, calculatedItemTotal 1275, totalMatches true, creditValid true, and transactionValid true. The reported updatedBalance is 2475 against a 5000 credit limit.

Validation interpretation

The staged order consists of two items: 2 x 450 = 900 and 5 x 75 = 375, totaling 1275. The simulated credit check computes 1200 + 1275 = 2475, below the 5000 limit. The output status COMMITTED is a workflow label, not proof of a database commit.

5. Evidence 03 - Credit-limit rejection

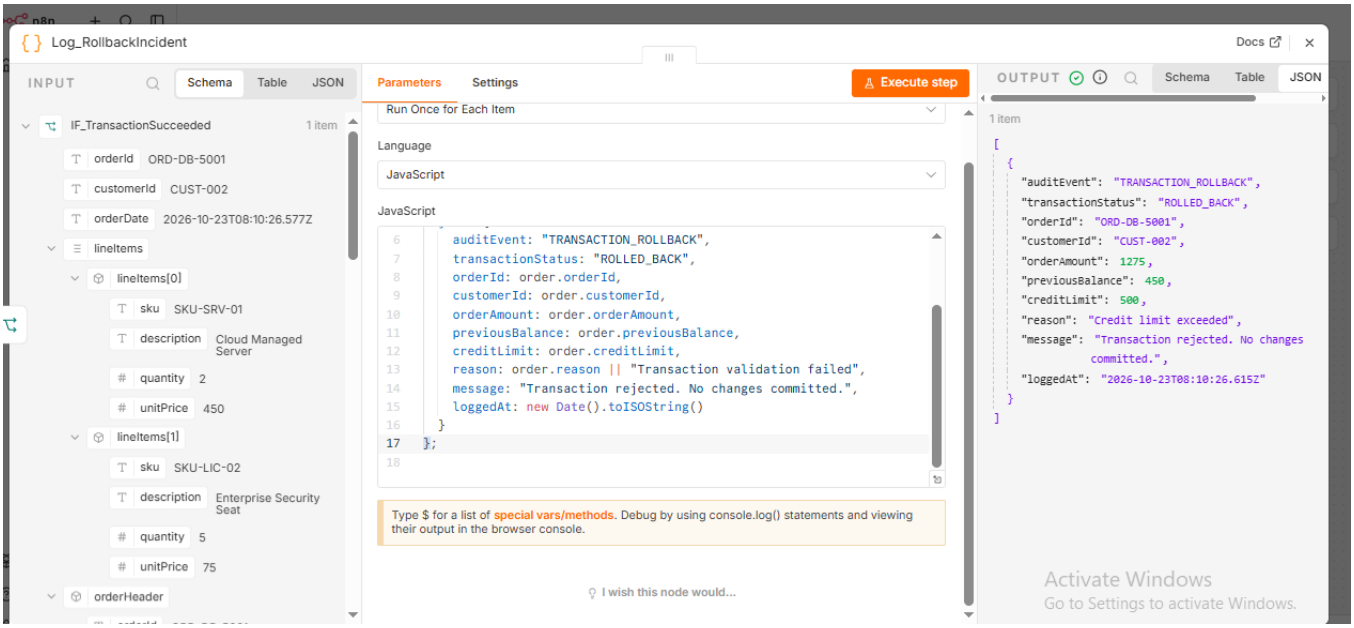


Figure 3. Rollback incident output for CUST-002 reports ROLLED_BACK, orderAmount 1275, previousBalance 450, creditLimit 500, and reason Credit limit exceeded. The output records the rejection; it does not prove a database rollback.

6. Test execution matrix

Test	Expected	Observed / evidence
T01 - Eligible customer	Accept; audit passes	PASS - Figure 2
T02 - Credit exceeded	Reject; log incident	PASS - Figure 3
T03 - Line item reconciliation	900 + 375 = 1275	PASS - Figure 2
T04 - IF branch routing	Success -> True; rejection -> False	PASS - Figures 1 and 3